

**To:** The Editor-in-Chief  
Journal of Process Control

**From:**

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**Re:** Submission of manuscript

*“Robust High-Order Control Barrier Function Safety Filtering for Supercritical Power Plant Control under Model Mismatch”*

Dear Editor,

We submit the enclosed manuscript for consideration at the Journal of Process Control. The paper introduces a Robust High-Order Control Barrier Function (Robust HOCBF) safety filter that combines Gaussian process mean correction with compositional uncertainty propagation through the HOCBF  $\psi$ -chain, providing a per-constraint, relative-degree-aware safety certificate for energy process control under model mismatch.

The contribution is a process-control-oriented safety filter: the GP-corrected drift reduces the residual perturbation, and a recursive  $\sigma$ -chain certifies the remaining uncertainty through a GP confidence-bound-based high-probability forward-invariance guarantee. The filter is policy-agnostic—compatible with LQR, MPC, or learned upstream controllers—and is validated on a 5th-order supercritical boiler-turbine benchmark with  $\Phi$ -scaled nonlinear dynamics and six CBF-enforceable constraints.

**Fit with Journal of Process Control.** The manuscript addresses constraint satisfaction in energy process control under model mismatch, a core concern of the journal’s scope. The Robust HOCBF filter is proposed as a complement to existing process control techniques (MPC, offset-free MPC, disturbance-observer-based control). The paper includes process-control-specific content: IAE performance metrics, sampled-data implementation analysis, QP feasibility diagnostics, deployment decision guidance, and discussion of practical limitations including scenario-specific GP calibration and the  $\Delta g = 0$  assumption.

**Key contributions:**

1. Compositional GP uncertainty propagation through the HOCBF  $\psi$ -chain, yielding a per-constraint, relative-degree-aware robustness margin with a formal high-probability safety certificate (Theorem 1).
2. Process-control-oriented safety certification with sampled-data implementation analysis, QP feasibility conditions, and one-sided diagnostic characterization.
3. Validation on a 5th-order supercritical boiler-turbine benchmark: nominal HOCBF fails catastrophically ( $> 95\%$  violation) under model mismatch; the Robust HOCBF achieves no observed CBF violations across six static perturbation scenarios with per-step QP latency  $\sim 25$  ms ( $10\times$  faster than the implemented SLSQP-based NMPC baseline).

**Conflicts of interest.** The authors declare no competing interests. The manuscript has not been published elsewhere and is not under simultaneous consideration by any other journal.

**Suggested reviewers:**

- Prof. Melanie N. Zeilinger, ETH Zurich, mzeilinger@ethz.ch (safe learning-based control, GP-MPC)
- Prof. Lukas Hewing, University of Oxford, lukas.hewing@eng.ox.ac.uk (cautious MPC, GP for control)
- Jianxin Zhou, Southeast University, zjx@seu.edu.cn
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We thank you for your time and consideration, and look forward to the reviewers' feedback.

Sincerely,

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